

MAC protocols, frame sharing (FRASH) MAC protocol, cluster based on demand multichannel MAC protocol, Q-MAC, RL-MAC and many more]. Energy efficiency can be obtained by implementing new algorithms on the physical layer (modulation techniques, architecture, etc) as well as on the MAC layer. Many image- and video-compression algorithms are also introduced, for instance, distributed video coding (DVC), high-efficiency video coding (HEVC), discrete cosine transform (DCT), discrete wavelet transform (DWT) and curvelet transform. Results show that by using advanced compression techniques, energy consumption can be reduced by up to 50 per cent.

Inherent uncertainty. In most WSNs, mobility of nodes creates problems as it affects topology, routing protocols and synchronisation. Coverage area and coherence bandwidth depend on moving obstacles and the surrounding environment.

Security and privacy. Multimedia industrial monitoring and control kind

of applications require high privacy and security. Unauthenticated persons should not be able to access private data.

Quality of service (QoS) and quality of information(QoI). High QoS and QoI are required on each and every communication layer. The resolution should be good enough at remote places. For video streaming, 5fps to 20fps (frames per second) is expected for better quality, and all layers should be designed accordingly.

Applications

WMSNs have numerous killer applications. Multimedia surveillance sensor networks can be used to enhance and complement existing surveillance systems to prevent crime and terrorist attacks. These can be used to locate missing persons, identify criminals or terrorists and record other potentially relevant activities, such as thefts, car accidents and traffic violations. With the help of WMSNs, it will be possible to monitor car traffic in big cities

and on highways that offer traffic-routing advice to prevent congestion or identify violations.

WMSNs play a vital role in personal health care units. Telemedicine sensor networks can be integrated with 3G and 4G mobile networks to provide ubiquitous health care services. Patients can carry various types of medical sensors to monitor different parameters, such as blood pressure, ECG, breathing activity, pulse oximetry and body temperature. Remote medical centres or personal doctors can monitor the condition of patients to infer emergency situations. This will be more beneficial for villages and rural areas where doctors and health centres are not available all the time.

WMSNs can also be used for environmental and structural monitoring purposes. Video and image sensors may be used to monitor the structural health of bridges and other civil structures. In industrial applications, WMSNs can give visual inspections and automated actions. ●



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